Chemical properties:

The chemical properties of oxygen and ozone differ vastly. Oxygen combines with many metals and non-metals to form oxides. With some elements such as s-block elements combination of oxygen occurs at room temperature. Some of less reactive metals react when powdered finely and made to react exothermically with oxygen at room temperature but a lump of metal is unaffected under same condition. These finely divided metals are known as pyrophoric and when set the powder on fire, heat is liberated during a reaction.

On the other hand ozone is a powerful oxidising agent and it reacts with many substances under conditions where oxygen will not react. For example, it oxidises potassium iodide to iodine. This reaction is quantitative and can be used for estimation of ozone.

$$O_3 + 2KI + H_2O \longrightarrow 2KOH + O_2 + I_2$$

Ozone is commonly used for oxidation of organic compounds. In acidic solution ozone exceeds the oxidising power of fluorine and atomic oxygen. The rate of decomposition of ozone drops sharply in alkaline solution.

Uses:

1. Oxygen is one of the essential component for the survival of living organisms.
2. It is used in welding (oxyacetylene welding)
3. Liquid oxygen is used as fuel in rockets etc...

3.2.1 Allotrophic forms of sulphur

Sulphur exists in crystalline as well as amorphous allotrophic forms. The crystalline form includes rhombic sulphur (α sulphur) and monoclinic sulphur (β sulphur). Amorphous allotropic form includes plastic sulphur (γ sulphur), milk of sulphur and colloidal sulphur.

Rhombic sulphur also known as α sulphur, is the only thermodynamically stable allotropic form at ordinary temperature and pressure. The crystals have a characteristic yellow colour and composed of S_8 molecules. When heated slowly above 96 ^{0}C, it converts into monoclinic sulphur. Upon cooling below 96 ^{0}C the β form converts back to α form. **Monoclinic sulphur** also contains S_8 molecules in addition to small amount of S_6 molecules. It exists as a long needle like prism and is also called as prismatic sulphur. It is stable between 96 ^{0} - 119 ^{0}C and slowly changes into rhombic sulphur.

When molten sulphur is poured into cold water a yellow rubbery ribbon of plastic sulphur is produced. They are very soft and can be stretched easily. On standing (cooling slowly) it slowly becomes hard and changes to stable rhombic sulphur.

Sulphur also exists in liquid and gaseous states. At around 140 ^{0}C the monoclinic sulphur melts to form mobile pale yellow liquid called λ sulphur. The vapour over the liquid sulphur consists of 90 % of S_8, S_7 & S_6 and small amount of mixture of S_2, S_3, S_4, S_5 molecules.

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